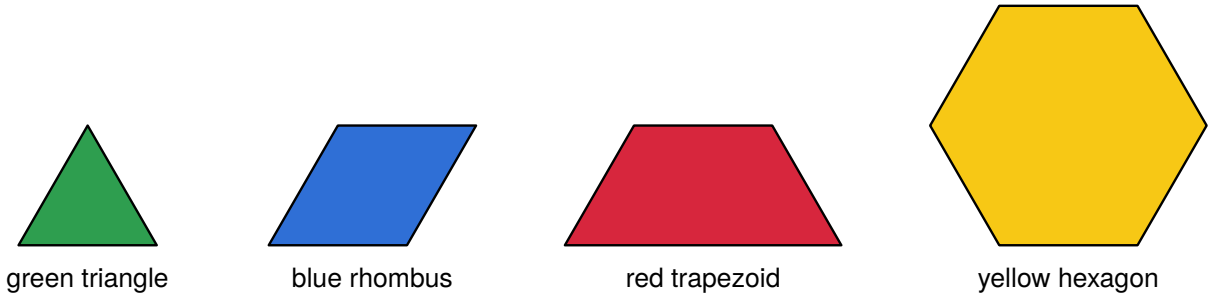
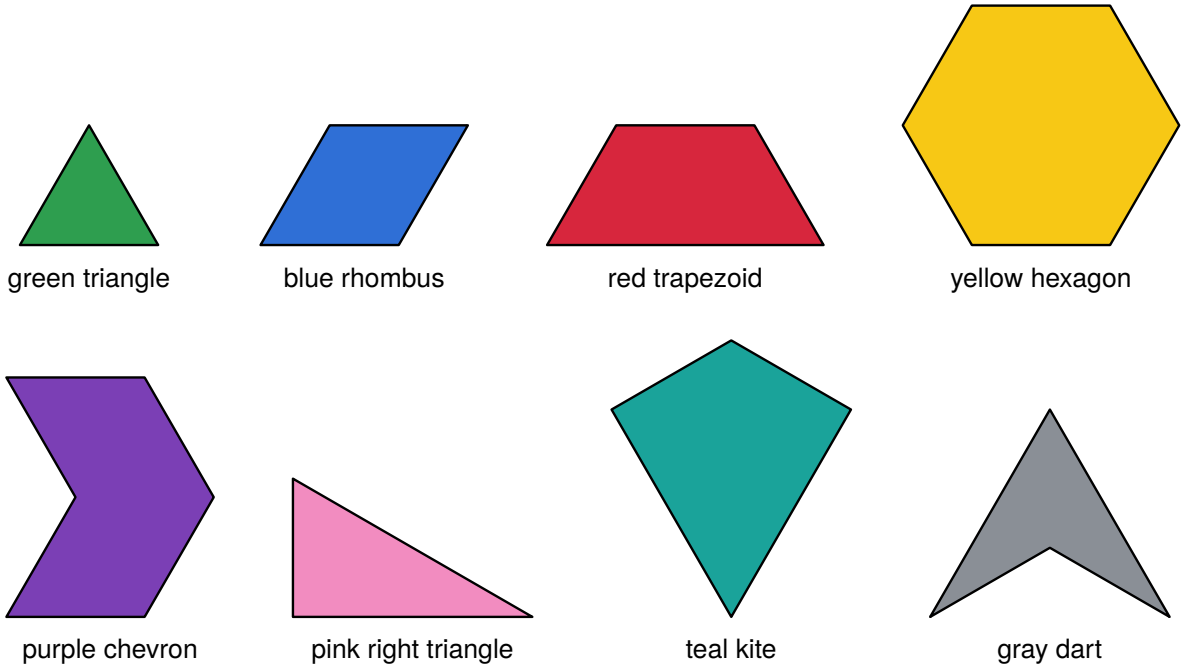
Problem 1: Your partner picks one of these four blocks in secret. Can you always find out which one with two yes-or-no questions?



green triangle | blue rhombus | red trapezoid | yellow hexagon

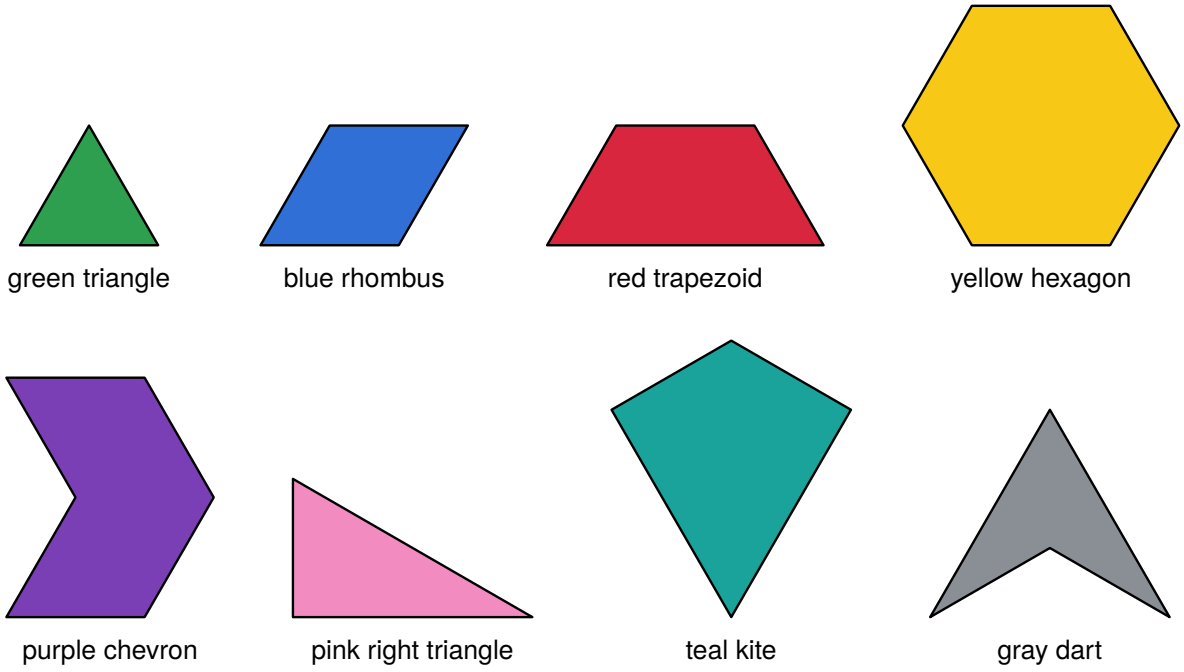
Problem 2: Your partner picks one of these eight blocks in secret. You can only point and ask “Is it this one?” What is the most questions you could need before you are sure?



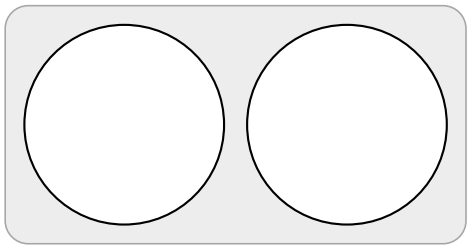
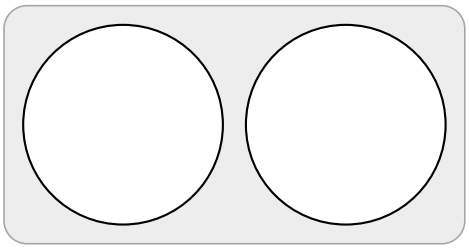
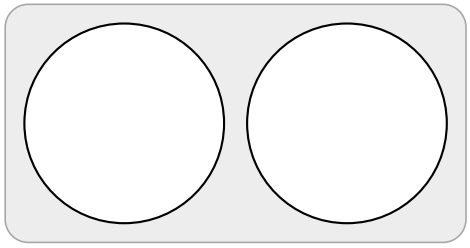
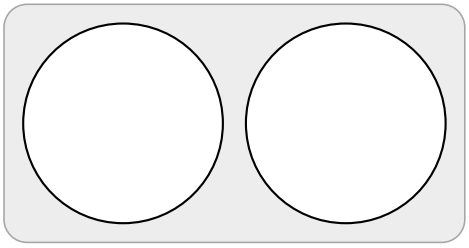
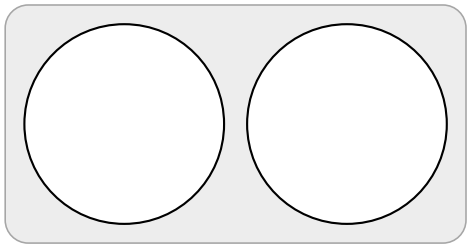
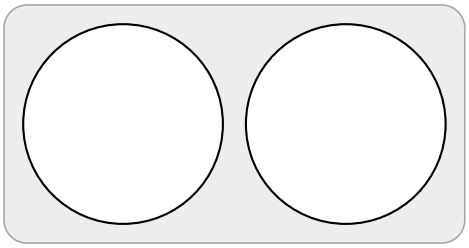
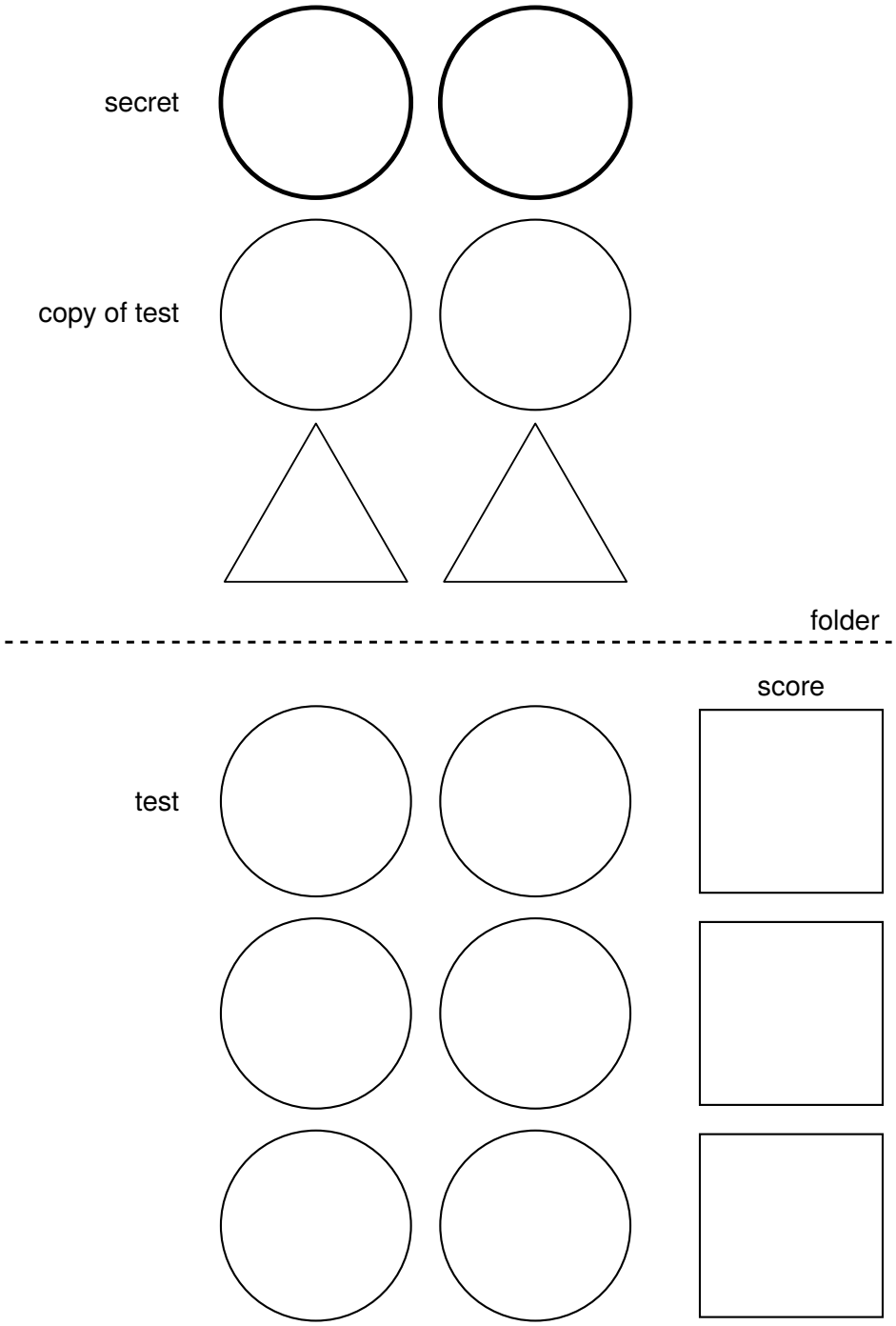
green triangle | blue rhombus | red trapezoid | yellow hexagon

purple chevron | pink right triangle | teal kite | gray dart

Problem 3: Your partner picks one of these eight blocks in secret. Can you always find out which one with three yes-or-no questions?



Problem 5: A secret is two counters in a row, each one red or yellow. How many different secrets can you make?








Problem 6: The grown-up hides two counters behind the folder. Put two counters in a test row to make a test. The grown-up puts a green triangle under each counter that matches the secret and tells you how many. When you say what the secret is, the grown-up lifts the folder and you check every score together. Can you always be sure of the secret after one test? Can you always be sure after two tests?